

MATTHEW GERGLEY

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EDUCATION

Utah Tech University

08/21/2022 – 05/03/2024

Bachelor of Science in Mathematics, cGPA: 3.88/4.0

St. George, UT

- Key Coursework: Abstract Algebra, Number Theory, Discrete Mathematics, Real Analysis, Statistical Inference, Mathematical Modeling, Physics I/II, Python, MatLab
- **Presidents List:** Spring 2023, Fall 2023, Spring 2024; **Deans List:** Fall 2022

University of Massachusetts Amherst

08/28/2021 – 05/10/2022

Major: Mathematics; Transferred after an academic year, cGPA: 2.98/4.0

Amherst, MA

- Key Coursework: Linear Algebra, Differential Equations, Statistics, Multivariate Calculus

TECHNICAL SKILLS

Languages: Python (Advanced), C++ (Intermediate), SQL (Beginner), MatLab (Intermediate)

Skills: Mathematics, Abstract Algebra, Number Theory, Linear Algebra, Mathematical Modeling, Classical Mechanics, Simulation, PCA/TOPSIS, ODEs, Statistics/Statistical Inference

Key Libraries: Pandas, NumPy, SciPy, GeoPy, Matplotlib, PyTorch

KEY RESEARCH PROJECTS / PRESENTATIONS

A Mathematical Model of HPA Axis Dynamics and Impacts of Alcohol Consumption **** Published ****

[Python, MatLab, Differential Equations]

06/07/2023 – 02/01/2026

- Developed a mathematical model utilizing a negative feedback loop showing how varying levels of alcohol consumption impacts stress response.
- Included circadian drive analysis relative to an individual's BAC.
- Presented at the **Joint Mathematics Meeting 2024** (JMM) in San Francisco, CA, the **Utah Tech Research Symposium 2024**, and the **International Mathematics and Statistics Student Research Symposium 2025**.
- Published Q1 in the IJMCE: <http://dx.doi.org/10.2478/ijmce-2026-0006>
- **10-15 hours per week**

Optimizing Police Patrolling

[Linear Algebra, Statistics, Optimization, Python, SQL]

01/10/2023 – 05/05/2023

- Worked with the Santa Clara/Ivins Police Department to develop a patrol route that optimizes response time and also ideal shift change times.
- Generated heat maps for 911 call locations based on a call severity scale and provided insights into seasonal differences.
- Presented to the **Santa Clara/Ivins Police Department** and at the **Santa Clara/Ivins City Alliance Luncheon**.
- **10-15 hours per week**

Maximizing Astronaut Productivity

[Differential Equations, Statistics, Optimization, Python]

08/21/2022 – 12/15/2022

- Developed astronaut work schedule, via our mathematical model, that maximizes astronaut productivity while minimizing stress.
- Utilizing a normal distribution to model productivity in relation to cortisol levels following the Yerkes-Dodson Law.
- Presented at the **MAA Intermountain Section Meeting 2023** and the **Utah Tech Research Symposium 2023**.
- **10 hours per week**

EXPERIENCE

Mathematics Tutor

09/12/2022 – 05/03/2024

Utah Tech University

St. George, UT

- Tutored university students across various mathematics disciplines, improving their understanding and academic performance.
- Collaborated with other tutors and managers in order to improve efficiency in the tutoring center.
- **15-18 hours per week**

SERVICES

PREP Summer STEM Program Assistant

05/15/2023 - 08/01/2023

Utah Tech University / AmeriCorps

St. George, UT

- AmeriCorps position.
- Ensured the safety and appropriate behavior of seventh-grade program participants.
- Assisted teachers in grading, hands-on activities, obtaining supplies, data collection, etc. in the classroom.
- Served as a mentor to students, encouraging the development of a commitment to educational achievement.
- **32-35 hours per week**

PERSONAL PROJECTS / SELF-STUDY

Pure Mathematics Self-Study

May 2024 - present

Self-Study

Harwinton, CT

- Documenting all my works in a GitHub website I created in HTML (<https://mgergley.github.io>)
- Algebra: studying and completing exercises from Algebra by Lang and Abstract Algebra by Saracino. Notes and proofs are typed in LaTeX and uploaded to my website.
- Differential Geometry: studying and completing exercises from Introduction to Smooth Manifolds by Lee. Notes and proofs are typed in LaTeX and uploaded to my website.
- Real Analysis: studying and completing exercises from Real Analysis by Cummings. Notes and proofs are typed in LaTeX and uploaded to my website.
- Topology: studying and completing exercises from Introduction to Topology by Gamelin & Greene
- Number Theory: studying and completing number-theoretic exercises from abstract algebra and online resources along with number theory related projects in Cryptography.
- **30-40 hours of study work per week**

Orbital Mechanics

02/26/2025 - 06/02/2025

Personal Project

Harwinton, CT

- Simulating the orbit of a satellite in low Earth orbit (LEO).
- Accounting for Earth's oblateness through J2 perturbation.
- Simulating and calculating Δv for plane change maneuver.
- Creating and maintaining working Python script and a LaTeX document outlining the mathematics/physics utilized and needed.
- **5 hours per week**